

# Chapter 163 — Focal Therapy of Prostate Cancer

Campbell-Walsh-Wein Urology · Chapter Notes

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- **Global incidence about 30.6 per 100,000 per year**; prostate cancer is the **second most incident cancer in men** worldwide.
- In the United States it is the **commonest solid-organ malignancy in men** and the **second leading cause of cancer death in men**.
- **Incidence has risen over a 10-year period**, particularly in countries with a **high Human Development Index and per-capita gross domestic product**.
- **Overdiagnosis and overtreatment controversy**: radical treatment is oncologically effective, but more men than necessary bear its **collateral damage to the cavernosal nerves, urinary sphincter and adjacent organs**.
- Three developments drive interest in **gland-sparing targeted therapy**: **multiparametric magnetic resonance imaging (mpMRI)**, **MRI-fusion targeted biopsy**, and recognition that **Gleason grade 4 and 5 cancers are the biologically lethal subgroup**.
- **Definition — focal therapy (FT)**: guided ablation of a cancerous lesion that is **image defined and biopsy confirmed (or biopsy confirmed only)**, with a **safety margin** around the target.

## Concepts and Historical Perspective

- **Model: breast conservation**. The Halsted mastectomy gave way to local excision with reconstruction; prostate FT followed that paradigm shift.
- Evolution: **subtotal ablation sparing one or both neurovascular bundles** → "male lumpectomy" (hemiablation), the first form of FT → true image-guided focal destruction with a margin of normal tissue.
- **Early ablations lacked advanced imaging**; localisation ranged from **standard transrectal biopsy to 5-mm transperineal template grids**.
- Treatment always balances **cancer aggressiveness against life expectancy**. FT aims to treat **only the foci that threaten survival or quality of life**, preserving surrounding tissue and so **sexual and urinary function**.

**Concept** — FT is the prostate's version of breast conservation: its legitimacy depends entirely on being able to see, sample and ablate the lesion that matters.

## Biologic Basis for Prostate Focal Therapy

### Using Cancer Grade as an Indication of Aggressiveness

- **Best predictors of prostate cancer mortality**: tumour grade, serum prostate-specific antigen (PSA) and clinical stage. In low-volume localised disease, **grade is the earliest indicator of lethal potential**.
- **Gleason grading (Donald Gleason, 1973)**, in modified form, remains the **most effective practical prognosticator** of aggressiveness.
- **Hanahan-Weinberg hallmarks of cancer (2000; expanded 2011)**: acquired self-sufficiency, immortality and invasive capability (Fig. 163.1). **Gleason grade 4 and 5 carry many more hallmarks than grade 3**.

**HALLMARKS OF MALIGNANCY IN GLEASON 4/5 VERSUS GLEASON 3**

| Hallmark                          | Molecular evidence favouring high grade                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Unopposed growth</b>           | Hundreds of genes differentially expressed between 3+3 and 4+4 (laser-capture microdissection): <b>androgen receptor, epidermal growth factor receptor (EGFR), tumour necrosis factor-<math>\alpha</math> (TNF-<math>\alpha</math>), Notch</b> , developmental and embryonal stem-cell pathways. EGFR copy number is <b>not</b> increased on fluorescence in situ hybridisation (FISH), but <b>aneusomy of chromosomes 7, 8 and 17 and HER-2/neu amplification occur almost only in pattern 4</b> . |
| <b>Loss of tumour suppression</b> | <b>Size of heterozygous 6q and 16q deletions</b> (the commonest deleted segments, rich in suppressor genes) <b>correlates with Gleason score</b> .                                                                                                                                                                                                                                                                                                                                                  |
| <b>Loss of cell-cycle control</b> | <b>Cyclin D2 promoter methylation</b> rises with Gleason score; <b>loss of p27(kip1)</b> correlates with grade.                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Resistance to apoptosis</b>    | An 86-gene panel separates pattern 3 from 4/5. <b>Defender against apoptosis 1 (DAD1)</b> , a nuclear factor- $\kappa$ B (NF- $\kappa$ B) target linked with perineural invasion, stains more intensely in grade 4/5. <b>Bcl-2 is overexpressed already in hormone-naïve high-grade, high-stage, highly proliferative cancer</b> .                                                                                                                                                                  |
| <b>Angiogenesis</b>               | <b>Small-calibre, irregular vessels</b> (CD34 immunostaining) mark men far more likely to die of cancer. <b>Vascular endothelial growth factor (VEGF)/PK-1 expression rises with Gleason score</b> .                                                                                                                                                                                                                                                                                                |
| <b>Invasion and metastasis</b>    | The <b>epithelial–mesenchymal transition (EMT) marker profile tracks Gleason score</b> . <b>Secreted frizzled-related protein 4 (SFRP4)</b> , a Wnt regulator, is higher in grade groups 3–5 and in lethal and metastatic cancer.                                                                                                                                                                                                                                                                   |

- **What the table is for:** every hallmark points the same way, so **pattern 4/5 is the priority target**. In early low-volume disease, focal treatment of these areas may eradicate lethal cancer while the **benign/pattern 3 remainder is surveilled**.
- **Trap:** EGFR copy number does **not** rise in pattern 4 on FISH. The pattern 4 abnormalities are chromosomal aneusomy and HER-2/neu amplification.

**Figure legends**

- **Fig. 163.1:** emerging and established hallmark capabilities and enabling characteristics of cancer (modified from Hanahan and Weinberg, 2011).
- **Fig. 163.2:** (A) unifocal cancer; (B) multifocal cancer with a clear index lesion and smaller secondary foci (the commonest pattern); (C) multifocal cancer with an unclear index tumour. Metastases to brain, lung, liver, spine, pelvic nodes and bone arise from the index lesion (from Perera et al., 2016).

**Multifocality of Prostate Cancer Versus the Index Lesion Hypothesis**

- **Most prostate cancer is multifocal (50–91%)** in radical prostatectomy specimens. One Korean series is the exception, reporting a unifocal majority (Table 163.1).
- **Index lesion (IL)** = the **largest or highest-grade tumour focus**. The **index lesion hypothesis** holds that the IL determines the course of the cancer.

**INDEX LESION HYPOTHESIS — EVIDENCE FOR AND AGAINST**

| Supports the hypothesis                                                                                                                                                                   | Challenges the hypothesis                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. The IL sets grade and stage:</b> the large majority of extracapsular extension arises from the IL, especially when PSA is <10 ng/mL.                                                | <b>Aggressive features (largest size, highest grade, highest stage) sit in a satellite rather than the IL 12–38% of the time.</b>                                                                                                            |
| <b>2. The IL carries most of the tumour volume,</b> roughly 80–90%. Total and index volume both predict biochemical recurrence, so treating the IL deals with most intraglandular tumour. | A minority of ILs have a <b>lower Gleason score than an adjacent nonindex lesion</b> , and extraprostatic extension (EPE) can arise from nonindex lesions. The IL Gleason matches the overall Gleason in only about two-thirds of specimens. |

| Supports the hypothesis                                                                                                                                                                                                                     | Challenges the hypothesis                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>3. Satellites are small and low grade. Stamey (1993):</b> insignificant cancer <0.5 mL. Unifocal cancer is 12.4–25%, but counting satellites <0.5 mL as insignificant raises "unifocal" to 52–93.4%. Most secondary cancers are <0.5 mL. | Yet in most multifocal glands (about 9 in 10), the <b>highest grade, largest volume and EPE do all lie in the IL.</b>   |
| <b>4. Metastases are monoclonal.</b> Multifocal lesions differ in mutation, copy number, methylation and expression (multiple lineages). At autopsy, metastases share the <b>copy-number signature of a single parent cell.</b>             | <b>Case report: a lethal metastatic clone traced to a small Gleason pattern 3 focus, not to the high-grade lesions.</b> |

- **Stamey threshold, derivation:** Surveillance, Epidemiology, and End Results (SEER) data put the lifetime probability of significant cancer at 8%. The largest 8% of cancers in the Stanford cystoprostatectomy series measured **0.5–6.1 mL**, so 0.5 mL became the cut-off.
- Applying Stamey's rationale to modern series gives **higher thresholds for significant cancer: >1.09 mL (Charing Cross) and >1.31 mL (Rotterdam arm of the European Randomised Study of Screening for Prostate Cancer).** So far more men would be eligible for FT than the older pathology suggests.
- **Conclusion:** if the IL is accurately identified and ablated, most of the tumour and its most aggressive components are treated. In theory, removing the lethal component could prevent metastasis and death.
- **These biological exceptions explain some FT failures, so close post-treatment surveillance is mandatory** for timely salvage. The same exceptions also cause failures of **whole-gland therapy and active surveillance (AS).**

TABLE 163.1 Whole-Mount Histology Studies Demonstrating the Multifocality of Prostate Cancer

Whole-mount radical prostatectomy series, 1992–2015; rows grouped by era.

| Era of publication                                                             | Organ-confined (pT2) | Unifocal                              | Unifocal once satellites <0.5 mL are discounted | Multifocal                             |
|--------------------------------------------------------------------------------|----------------------|---------------------------------------|-------------------------------------------------|----------------------------------------|
| <b>1992–2000</b> (5 series)                                                    | Rarely reported      | <b>17–50%</b>                         | Not reported                                    | <b>50–83%</b>                          |
| <b>2002–2015</b> , PSA-screening era (14 series; PSA means/medians 3–16 ng/mL) | 50–89%               | <b>8.9–36%</b> (Korean outlier 66.7%) | <b>33–93.4%</b>                                 | <b>64–99.1%</b> (Korean outlier 33.3%) |

- **What the table is for:** it shows that **multifocality is the rule**, which looks like the central objection to FT.
- **Pattern across the rows:** multifocality is higher in the PSA era. Once small satellites are discounted, however, most glands become **functionally unifocal**, and that turns the table from an objection into an argument for FT.
- **Range in the text vs the table:** the text quotes **multifocality 50–91%**, while the table runs from **33.3% (Korean series) to 99.1%**. The text range sets aside the extremes. For "unifocal with satellites", the text gives 52–93.4% and the table 33–93.4%.
- **Trap:** "multifocal, therefore unsuitable for FT." What matters is whether there is **one significant lesion plus insignificant satellites**, not the count of foci.

**Concept** — The index lesion hypothesis turns prostate cancer **from a disease of the gland into a disease of one lesion**. FT stands or falls on it, and its exceptions are why FT is always paired with surveillance.

## CLINICAL APPLICATIONS OF FOCAL THERAPY

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### Treating the Index Lesion

- It is increasingly accepted that the natural history is **driven by a primary lesion rather than by secondary lesions**.
- Eradicating the dangerous foci gives **oncological control while better preserving quality of life**. The **secondary lesions are placed on AS**.

### The Complementary Roles of Focal Therapy and Active Surveillance

- The two largest AS cohorts now have **10–15 years of follow-up** and show an **extremely low metastasis rate in D'Amico low-risk disease**.
- In the more inclusive cohort (up to low-volume 3+4), **most men who progressed had Gleason 3+4**.
- **Prostate Intervention Versus Observation Trial (PIVOT)**: the survival benefit of prostatectomy over observation was seen **in the intermediate-risk subgroup**.
- Hence **grade 4 is the therapeutic target**. The strategy is **FT to intermediate- or high-grade foci plus AS for the low-grade remainder**, with focal retreatment of any new higher-grade focus. The aim is to **turn prostate cancer into a chronic disease**, in keeping with its long natural history.
- **Multidisciplinary team sequence** (uro-oncologists, radiologists, pathologists): **map disease extent by advanced imaging and biopsy → confirm benefit and compliance with follow-up → completely ablate the IL(s) → monitor with imaging and biopsy → retreat persistent or de novo disease focally, or convert to whole-gland treatment**.

*Concept — FT and AS are not rivals but **two halves of one strategy**: ablate the grade 4 disease that kills, and watch the grade 3 disease that does not.*

## PATIENT SELECTION

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- **Patient selection is the fundamental tenet of FT**. The aggressive lesion must be identified and its **extent delineated for complete ablation**.
- Recurrence and persistence are more likely than after radical therapy, so the patient must be **compliant, willing and psychologically suited to close surveillance**.
- The lesion is identified **histologically and/or radiologically**. mpMRI shows the whole gland and may predict aggressive histology, but **consensus requires biopsy confirmation**.
- Imaging is imperfect at **excluding** aggressive cancer, so **many experts add a systematic biopsy**.

### Biopsy-Based Lesion Identification

- Early work aimed at **lateralisation for hemiablation**, largely in low-grade disease that would now go to AS. It still shows **how well biopsy maps cancer**.
- **Transrectal ultrasonography (TRUS)-guided biopsy unilaterality is the strongest predictor of pathological unilaterality**. Sensitivity and **negative predictive value (NPV) are high**, but **specificity and positive predictive value (PPV) are poor**.
- **Extended cores beat sextant cores** on specificity and NPV, but even **24-core TRUS leaves specificity low**. Adding transrectal cores has a **ceiling** because sampling is random and the **anterior zone is poorly accessible** transrectally.
- **Transperineal mapping biopsy (TMB), 1990s**: **equal access to anterior and posterior gland**, and a **5-mm grid or template** gives consistent core density and distribution.

- Against TMB, **staging TRUS biopsy misses almost half** of the disease relevant to hemiablation. **More than half of men with unilateral cancer on TRUS are disqualified after TMB.**
- **Predictive models are weak.** Only **PSA  $\leq 4$  ng/mL and unilateral cancer on biopsy** predicted unilateral disease; another model reached only chance-level discrimination (area under the curve, AUC, 0.52). **More than 2 positive TRUS cores** predicted persistence or biochemical recurrence after focal treatment.
- These data aimed to locate **all** cancer. Current interest is in **preferential detection of Gleason 4/5**, because detecting more pattern 3 disease drives overtreatment.

#### TRANSRECTAL VERSUS TRANSPERINEAL BIOPSY FOR FOCAL THERAPY MAPPING

|                   | Systematic TRUS biopsy                                 | Transperineal mapping biopsy                  |
|-------------------|--------------------------------------------------------|-----------------------------------------------|
| Anterior zone     | Poorly sampled; no standard method                     | Equal anterior and posterior access           |
| Core distribution | Random                                                 | 5-mm grid or template, consistent             |
| Lateralisation    | High NPV, poor specificity; ceiling despite more cores | Reference standard of the pre-MRI era         |
| Consequence       | Over-selects men as "unilateral"                       | Disqualifies over half of TRUS-unilateral men |

### Advanced Imaging Techniques in Prostate Cancer

#### Multiparametric MRI (mpMRI)

- **High-field magnets, contrast agents and better protocols** added functional to anatomical imaging, giving **high diagnostic accuracy**. mpMRI is the **key enabler of FT**. It can do the following:
  1. **Preferentially detect high-grade lesions.** Dynamic contrast-enhanced (DCE) gadolinium imaging picks up vascularised, **larger, higher-grade foci**. Diffusion-weighted imaging (DWI) maps water mobility and shows the **high cellularity of lost glandular architecture**.
  2. **Detect anterior-zone cancer**, which is **15–33% of cancers**. These tumours are **diagnosed late because TRUS biopsy misses them**, but are readily seen on mpMRI. Caught early, they are **ideal for FT: complete ablation sparing both neurovascular bundles**.
  3. **Detect EPE**: accuracy 62–74%, sensitivity 35–63%, specificity 73–91%, PPV 36–85%, NPV 57–92%. mpMRI **outperforms the Partin tables** and adds limited pelvic staging.
  4. **Identify the IL**. Sensitivity is modest overall but **highest for tumours >1 cm, Gleason  $\geq 7$  and index tumours**. IL location agrees closely with the prostatectomy specimen. Significant cancers that mpMRI misses are mostly **nonindex foci with <20% pattern 4**.
- **PROMIS** (mpMRI against 5-mm grid TMB, 11 centres with varied equipment and expertise) for clinically significant prostate cancer (csPCa): **sensitivity 93%, specificity 41%, PPV 51%, NPV 89%**.
- **PROMIS implications**: mpMRI lesions need **histological confirmation**; **systematic biopsy of MRI-negative areas cannot yet be omitted**; and a strategy relying heavily on mpMRI risks a **small but real amount of outfield disease**.
- **Technical dependence**: **3 tesla is more accurate than 1.5 tesla**, and an **endorectal coil is recommended at 1.5 tesla**. Protocol matters too. **Reader experience** determines IL identification and EPE interpretation.
- **mpMRI underestimates histological tumour volume by 5–30%** (after allowing for shrinkage) and underestimates the axial diameter.
- **Treatment margin** (counting only pattern 4/5): **5 mm ablates >95%, 6 mm ablates 98.6%** of csPCa. Lesion-only ablation is therefore hard, and **complete ablation may require hemi- or zonal ablation**.

## MRI-Targeted/MRI-Fusion Biopsy

- mpMRI has **high sensitivity but low specificity**, so **histological confirmation of suspicious lesions is critical**.
- **Prostate Imaging-Reporting and Data System (PI-RADS)**: from the American College of Radiology and the European Society of Urogenital Radiology; now **version 2 (2015)**; a **5-point Likert scale** for csPCa. **Higher PI-RADS scores go with higher Gleason scores**.
- Histological Gleason score is still the decision marker. **2016 consensus: Gleason 3+4 is the "sweet spot" for FT**. Higher grades raise concern about **EPE outside the image-guided target zone**, which may need radical treatment.

### MRI-TARGETED BIOPSY TECHNIQUES COMPARED

| Technique                                             | Method                                                                                                                                                                   | Advantages                                               | Limitations                                                               |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------|
| <b>In-bore MRI-guided</b>                             | Live MRI guidance within the gantry                                                                                                                                      | <b>Highest accuracy</b>                                  | Slow, costly, MRI-compatible equipment; academic centres only             |
| <b>MRI-ultrasound software fusion</b>                 | <b>MRI segmentation → ultrasound segmentation → registration (rigid or elastic) → navigation</b> (mechanical, visual or combined) → biopsy; transrectal or transperineal | Standard ultrasound workflow; several commercial systems | <b>Imperfect overlay</b> ; operator must compensate; learning curve       |
| <b>Cognitive fusion</b> (visual estimation targeting) | Operator estimates lesion position from mpMRI                                                                                                                            | No extra equipment                                       | <b>Expert dependent</b> ; as good as software fusion only in expert hands |

- **MRI-targeted transperineal vs transrectal biopsy** in the same patients: **no difference in csPCa detection**.
- Tumours are identified accurately for FT **only by an experienced multidisciplinary team** of radiologists and urologists that **climbs the learning curve together**.

### Role of the Systematic Biopsy

- **Experts agree that a concurrent systematic biopsy is necessary** to find MRI-invisible csPCa. The **required intensity is disputed**.
- Reducing saturation biopsy (about 22 cores) to fewer systematic cores **progressively lowers csPCa detection** and **changes the FT planning template**.
- **About 90% of outfield csPCa lies within 1 cm of the target** (the lesion's "penumbra"), so biopsies should be taken **within and around** the MRI lesion.
- With limited systematic biopsy there is a **small but significant risk of untreated csPCa**. This affects biopsy-based FT outcomes, but the **long-term impact is unclear**.

**Concept** — *Imaging finds the lesion, but **only biopsy can grade it, and only a systematic biopsy can vouch for the rest of the gland**. That is why every consensus has kept both.*

## Promising Imaging Modalities

### Positron Emission Tomography

- **Positron emission tomography (PET)** with choline tracers or fluorodeoxyglucose has been investigated for staging.
- **Prostate-specific membrane antigen (PSMA)**: a **type 2 transmembrane protein expressed 100- to 1000-fold more** in prostate cancer than in normal cells.
- The early antibody agent <sup>111</sup>In-capromab pendetide **failed** to localise cancer reliably for quadrant ablation.

- **Small-molecule ligands to the external epitope**, labelled with  $^{18}\text{F}$  or  $^{68}\text{Ga}$ , give **high tumour uptake and rapid clearance**, so contrast is high.
- **PSMA PET is not degraded by biopsy haemorrhage or inflammation**, whereas mpMRI is transiently impaired.
- **mpMRI has greater sensitivity; PSMA PET ( $^{18}\text{F}$ -DCFBC) has greater specificity**, including for Gleason 8–9 disease.
- **Choline-PET and mpMRI targets can both be fused to three-dimensional (3D) TRUS**, combining anatomical and functional information for biopsy.

### Multiparametric Ultrasound

- Rationale: mpMRI is **resource intensive** and fusion is imperfect. Ultrasound is **real-time and point-of-care**, and a multiparametric ultrasound approach **avoids MRI-ultrasound registration error**.

### ULTRASOUND MODALITIES UNDER DEVELOPMENT

| Modality                                | Principle                                                                                                                                                  |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contrast-enhanced TRUS (CE-TRUS)        | Echogenic <b>microbubbles small enough to traverse capillaries</b> map microvascular distribution                                                          |
| TRUS elastography                       | <b>Tissue stiffness</b> from the properties of a reflected mechanical wave                                                                                 |
| Shear-wave elastography                 | Wave generated by the probe itself, so <b>reproducible and operator independent</b> ; <b>50 kPa cut-off</b> for cancer detection                           |
| Acoustic radiation force impulse (ARFI) | <b>Synchronised waves remove interference</b> from internal reflections; correlates well with mpMRI                                                        |
| Computerised TRUS (C-TRUS)              | Grey-scale image converted to <b>digital signals for statistical analysis</b> (1990s); basis for artificial-intelligence-driven multiparametric ultrasound |

- **Combining modalities adds functional information and raises individual sensitivities by 13–59%**. Prospective work is needed.

### Contemporary Patient Selection Criteria

- Criteria have evolved with biological understanding. **Early FT cohorts included low-risk men now managed by AS**.
- Treatment choice is patient-centric, so there are **few comparative trials between modalities**. Practice rests on **consensus** (Table 163.2).
- **Evolution of the stated aim**: 2009 — eradicate all known cancer and preserve genitourinary function → 2010 — treat cancer, leaving the benign prostate and surrounding structures intact → 2013 — treat the dominant lesion or IL; acceptable residual disease first quantified as **untreated 3+3 up to 5 mm core length** (biopsy type and intensity not addressed) → 2016 — **completely ablatable 3+4 is the "sweet spot"**; 4+3 the **maximum grade**; most panellists reluctant to treat high grade → 2019 — **treatment and retreatment to delay or avoid radical therapy**.
- **Assessment evolution**: mpMRI moved from "**state-of-the-art centres only**" to "**whenever possible**" (2016). TMB was **state of the art** in 2010 and 2012; MRI-TRUS fusion biopsy was **recommended by 2016**. The latest consensus **still requires systematic biopsy of the MRI-"normal" gland** because mpMRI has a small but significant false-negative rate.
- **Sexual function** (2013 trial-design consensus and 2016 consensus): **erectile dysfunction is not a contraindication to FT**. Potential preservation of erections is an important reason to choose FT, but **most experts would not deny FT to an impotent man**, given the urinary, rectal and other toxicities of indiscriminate whole-gland therapy.

- Selection remains an interplay of **disease and patient factors**. **Poor selection carries the cost and morbidity of treating recurrence.**

**TABLE 163.2 Summary of Contemporary Consensus on Patient Selection**

Consensus meetings 2009–2019, published as de la Rosette et al. (2010), Ahmed et al. (2012a), Donaldson et al. (2015), Tay et al. (2017a) and Gross et al. (2019).

| Variable                             | 2009 meeting<br>(publ. 2010)         | 2010 meeting<br>(publ. 2012) | 2013 meeting<br>(publ. 2015)                                 | 2016 meeting<br>(publ. 2017)                                                             | 2019<br>(publ. 2019)                                                                                        |
|--------------------------------------|--------------------------------------|------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Goal</b>                          | Eradicate all cancer                 | Eradicate all cancer         | <b>Eradicate csPCa</b>                                       | Eradicate csPCa                                                                          | Ablate completely;<br><b>delay or avoid radical therapy</b><br>through treatment, retreatment, surveillance |
| <b>Overall work-up</b>               | TMB                                  | TMB + mpMRI                  | TMB or systematic + mpMRI                                    | <b>mpMRI + systematic biopsy</b>                                                         | <b>mpMRI + targeted + systematic biopsy</b>                                                                 |
| <b>mpMRI</b>                         | State-of-the-art centres only        | Recommended                  | No clear recommendation                                      | Recommended as far as possible                                                           | Recommended                                                                                                 |
| <b>Biopsy of suspicious lesion</b>   | —                                    | To be studied                | Recommended                                                  | Recommended; <b>MRI-TRUS fusion</b>                                                      | Recommended                                                                                                 |
| <b>Biopsy of nonsuspicious gland</b> | —                                    | TMB                          | Recommended                                                  | Recommended; <b>≥12-core TRUS</b>                                                        | Recommended                                                                                                 |
| <b>Risk group</b>                    | Low to moderate                      | —                            | Intermediate                                                 | Intermediate                                                                             | Intermediate                                                                                                |
| <b>Maximum grade</b>                 | —                                    | —                            | —                                                            | <b>4+3</b>                                                                               | —, but <b>no cribriform or ductal</b>                                                                       |
| <b>Maximum size</b>                  | T2a clinically or T2b radiologically | —                            | No agreement                                                 | <b>1.5 mL or 3 mL if not crossing midline; 20% or 25% of prostate volume if crossing</b> | —                                                                                                           |
| <b>Residual disease permitted</b>    | None                                 | None                         | <b>3+3, core length ≤5 mm</b>                                | <b>3+3, core length ≤1 mm</b>                                                            | 3+3                                                                                                         |
| <b>Age/life expectancy</b>           | ≥10 years                            | —                            | Not primary; <b>best if &gt;10 years and not &lt;5 years</b> | Per major guidelines; <b>no upper or lower limit</b>                                     | —                                                                                                           |
| <b>Sexual function</b>               | Effect unknown; counsel              | —                            | —                                                            | <b>Important, but not the only indication</b>                                            | —                                                                                                           |

- **What the table is for:** tracking how **FT's target and toolkit have shifted**. The target moved from all cancer to csPCa to a retreatment strategy. Work-up moved from TMB to **mpMRI plus targeted plus systematic biopsy**.
- **Pattern:** the risk group settled on **intermediate**. **Systematic biopsy of the rest of the gland survives every iteration**.
- **Year labelling:** the text dates the consensuses by meeting (2009, 2010, 2013, 2016), the source table by publication (2010, 2012, 2015, 2017). Both are shown above.

- **mpMRI "state-of-the-art centres only"**: the text attributes this to the 2010 meeting, but the source table places it in the 2009-meeting column and marks the 2010 meeting "recommended". Both are recorded.
- **Trap**: tolerance of residual 3+3 **tightened** from 5 mm (2013) to **1 mm** (2016); it did not loosen.
- **Trap**: the size limit is **1.5 mL or 3 mL** if the lesion does not cross the midline. If it does, the limit is a **proportion of gland volume (20–25%)**.

**Concept** — Over successive consensuses the target of FT **narrowed from "all cancer" to "clinically significant cancer"** and its toolkit moved from template mapping to MRI with fusion. **The systematic biopsy has never been dropped.**

## ABLATION PATTERNS AND CURRENT TECHNOLOGIES

### Types of Ablation

- Many ablation patterns have been called FT, which makes the literature **heterogeneous**. **"Hockey-stick" ablations belong under partial prostate ablation**, not FT (Fig. 163.3).
- **International consensus on definitions: FT = lesion ablation, quadrant ablation and hemiablation.**
- Lesion treatment is further split into **image-targeted lesion ablation** (image-detected, biopsy-confirmed) and **non-image-targeted ablation** (biopsy information only).

### Figure legends

- **Fig. 163.3 — patterns of ablation: targeted, quadrant, anterior, hemiablation, hockey stick** (tumour focus inside a surrounding ablation zone).
- **Fig. 163.4 — focal cryoablation**, anterior ablation pattern (Tay and Polascik, 2017): see the sequence under Focal Cryotherapy.

### Ablation Approach

- Routes: **transrectal, transperineal, or (less commonly) transurethral**. Choice depends on **tumour location, available technology, physician experience and patient anatomy or preference**.
- **Transrectal gives easy access to the posterior zone; transperineal gives better access to the anterior zone.**

### ENERGY SOURCE AND DELIVERY ROUTE

| Energy                                   | Usual route                               | Reason                                                                                  |
|------------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------|
| Cryotherapy                              | Transperineal                             | Metal probes of 17 gauge or larger; poorly suited to transrectal use                    |
| Irreversible electroporation             | Transperineal                             | Same: rigid metal electrodes                                                            |
| High-intensity focused ultrasound (HIFU) | Transrectal (transurethral also possible) | Transducer housing must hold enough rings for the required intensity and focal distance |
| Laser                                    | Transrectal or transperineal              | Highly flexible fibres                                                                  |
| Transurethral devices                    | Transurethral                             | Deliver <b>HIFU or steam vapour</b>                                                     |

- As urologists master several techniques, an **"à la carte" approach** matches modality and route to the patient and lesion.

### Focal Cryotherapy

- **Thermal ablation by extraction of heat to lethal cold**. Cell death results from **intra- and extracellular ice** (membrane disruption), **recrystallisation, vascular stasis and ischaemia, apoptosis and immune-mediated effects**.

- Four developments make cryotherapy suited to FT:
  1. **Third-generation cryoprobes using the Joule–Thomson effect**, as fine as **17 gauge**.
  2. **Variable ice-length adjustment**, so the ice ball is contoured to the lesion.
  3. **Ice edge visible on TRUS, computed tomography (CT) or MRI**, giving real-time monitoring of the ablation zone and margin.
  4. **Urethral warmer**, which has greatly reduced urethral cold injury.
- **Thermocouples** are placed in the ablation zone or its margin and along **vital structures: external sphincter, neurovascular bundles, Denonvilliers fascia**.
- **Anterior focal cryoablation sequence** (Fig. 163.4): **biplanar ultrasound probe in the rectum → variable-length cryoprobes set to prostate measurements → four probes placed in the anterior prostate → ice formed, with a hyperechoic leading edge monitored → console confirms lethal temperature at the anterior apex and safe temperatures at the external sphincter and right neurovascular bundle**.
- **Series from 2008 to 2023** (Table 163.3) mostly used a **hemiablation template**, the original model of FT.
- Over a median follow-up of **6–85 months: biochemical progression-free survival (BPFS) 71–98%; 3-year BPFS 56–62.5%; 5-year BPFS 62–70%**.
- In cohorts mandating rebiopsy at 6–12 months, the **positive biopsy rate was 7.1–38.9%, about a quarter of it clinically significant**.
- Selection and biopsy criteria varied. Most patients were **D'Amico low or intermediate risk**.
- **Pad-free continence 95.1–100%; potency 46.8–100%**. About **half of men potent after hemiablation use phosphodiesterase type 5 inhibitors (PDE5I)**.
- **Clavien–Dindo grade ≥3 complications ≤4%** in every series. **Urinary retention is the commonest adverse event** (reactive oedema), **avoided by keeping a urethral catheter for a few weeks**. Retention 0–9%; **recto-urethral fistula (RUF) up to 0.8%**. Others: haematuria, haemospermia, urethral sloughing, stricture.
- **No comparison with radical prostatectomy or radiotherapy exists**. The comparisons are partial vs whole-gland cryotherapy in the **Cryo On-Line Data (COLD) registry**:
  - **Low risk (matched pairs): no difference in 5-year BPFS; erectile recovery significantly better with partial ablation**.
  - **Intermediate risk (propensity matched): no difference in early BPFS; partial ablation roughly twice as likely to restore erections at 12 months**. This reflects contemporary selection.

**TABLE 163.3 Summary of Published Focal Cryotherapy Series**

Series 2008–2023, including the COLD registry; rows grouped by selection era.

| Endpoint                   | 2008–2018: TRUS or TMB selection; mostly hemiablation; low and intermediate risk | 2019–2023: mpMRI + targeted/systematic biopsy; focal or targeted; 3+4 and 4+3 common |
|----------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Inclusion</b>           | Unilateral cancer; Gleason ≤7; PSA <10–20 ng/mL; ≤cT2b                           | MRI-visible, grade group 2–3, PSA <20 ng/mL; lesion <3 mL (single) or 1.5 mL (dual)  |
| <b>Median follow-up</b>    | 13–85 months                                                                     | 6–85 months                                                                          |
| <b>Biochemical control</b> | BPFS 56–98% (3-year 56–62.5%; 5-year 70–71.3%)                                   | Biochemical recurrence 25–30.6%; 3-year failure-free survival 90.5%                  |
| <b>Positive biopsy</b>     | Infield 1.4–12.5%; outfield 8.3–15.1%; overall 2.1–19.4%                         | <b>Infield 0–57%; outfield 4–43%</b> ; csPCa 21% at 12–18 months                     |
| <b>Sexual function</b>     | Ability to penetrate 46.8–90%; small IIEF fall, recovering                       | Ability to penetrate 54–84%; mild fall in IIEF or EPIC                               |
| <b>Urinary function</b>    | Continence 95.1–100%                                                             | Continence 96–100%                                                                   |

| Endpoint      | 2008–2018: TRUS or TMB selection; mostly hemiablation; low and intermediate risk | 2019–2023: mpMRI + targeted/systematic biopsy; focal or targeted; 3+4 and 4+3 common                                           |
|---------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Complications | Retention 1.2–15%; RUF 0–2%; cavernosal necrosis, urethral stenosis              | Retention up to about 30%; urinary tract infection up to about 25%; RUF rare; 43% needed a catheter up to 5 days in one series |

- **What the table is for:** cryotherapy has the **largest and longest FT literature**, and the only registry comparison.
- **Pattern:** as selection moved to mpMRI and higher grade, and as follow-up biopsy became **mandatory, targeted and systematic**, reported **positive biopsy rates rose**. Later series look for failure harder and treat more aggressive disease.
- **Text and table differ.** The text gives **retention 0–9% and RUF  $\leq 0.8\%$** ; the table includes retention of 15% and about 31% (37 of 119) and RUF of 2%. The text range covers the series reporting percentages, while later series report counts. For 5-year BDFS, the text gives 62–70% and the table 70–71.3%, because the text's lower bound comes from a different data cut.
- **Trap:** BDFS figures are **not comparable across series**, because different definitions were used (Phoenix, the American Society for Radiation Oncology (ASTRO) definition, nadir + 0.5) and none is validated after FT.
- Abbreviations: **IIEF**, International Index of Erectile Function; **EPIC**, Expanded Prostate Cancer Index Composite.

### Focal Salvage Cryotherapy

- **Local recurrence after definitive radiotherapy** is hard to manage: salvage must achieve **cancer control and limit salvage morbidity**.
- Focal salvage cryotherapy: **about half are biochemically free at 5 years, with slow further decline to 10 years**, and a favourable side-effect profile.
- **Focal and whole-gland salvage ablation give similar oncological control; focal is better functionally.** In the largest cohort, about three-quarters were free of biochemical progression at 2 years, and about half of initially potent men kept erections.
- **The most feared complication of salvage therapy is RUF:** about 1.4% in the largest cohort (Table 163.4).

**TABLE 163.4 Summary of Published Focal Salvage Cryotherapy Series**

Eight series, 2007–2021.

| Feature           | Range across series                                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Selection         | <b>Biopsy-proven radiorecurrent, nonmetastatic</b> cancer; staged by bone scan, MRI or CT, TRUS biopsy or TMB                          |
| Ablation plan     | Focal, partial or hemiablation                                                                                                         |
| Median follow-up  | 12–47 months                                                                                                                           |
| BDFS              | 1-year 51–62.6%; <b>2-year 72–80%</b> ; <b>5-year 46.5–54.4%</b> ; 10-year 42%                                                         |
| Other oncological | 3-year failure-free survival 40%, metastasis-free survival 50% (smallest series)                                                       |
| Erectile function | Ability to penetrate <b>14–50%</b> of men potent before salvage                                                                        |
| Continence        | 87–100%                                                                                                                                |
| Complications     | <b>RUF 1–3 cases per series (about 1.4%)</b> ; retention; lower urinary tract symptoms; urethral sloughing; bladder outlet obstruction |

- **What the table is for:** salvage outcomes are **roughly half of primary FT** for both cancer control and erections. **RUF is the complication that defines salvage risk**.

- **Trap:** expecting primary-FT functional results in the irradiated gland.

## Focal High-Intensity Focused Ultrasound

- A transducer focuses multiple ultrasound beams on a preset point, reaching  $\geq 55^{\circ}\text{C}$  → **coagulative necrosis**. The transducer sits in the **transrectal probe, which also plans and monitors treatment**.
- **Posterior-zone lesions are well suited. Anterior lesions are harder:** energy dissipates across intervening tissue, and anterior targets are displaced as progressive oedema develops.
- **Device limits: prostate size, height, and calcification in the treatment path.** Some recommend a **transurethral resection of the prostate (TURP) before HIFU** to reduce height and aid targeting.
- Whole-gland HIFU is slow; **focal treatment is shorter, so HIFU suits FT.**
- **Early biopsy-guided hemiablation** (pre-MRI, low risk): recurrence-free survival was high at 5 years but fell steeply by 10 years, with a high positive biopsy rate.
- **Series 2011–2022** (Table 163.5): mostly **hemiblativ**, using **cognitive or software MRI fusion**; mostly D'Amico low and intermediate risk. Median follow-up 6–45 months:
  - **Infield positive biopsy 16.7–54% (any cancer), 0–21.8% (csPCa)**
  - **Outfield positive biopsy 7–42.8% (any cancer), 2–7.5% (csPCa)**
  - **Continence 90–95%;** potency 50–78% at 6–12 months (heterogeneous, often non-validated definitions)
- **HEAT registry** (largest prospective focal HIFU study, 1379 men): **failure-free survival 68% intermediate risk and 65% high risk.**

**TABLE 163.5 Summary of Published Focal HIFU Series**

Series 2011–2022, including the HEAT registry (Reddy et al., 2022).

| Endpoint                | 2011–2012: biopsy-only selection                                                                       | 2015–2022: mpMRI + targeted, systematic or TMB                                                                                                          | HEAT registry (2022)                                                     |
|-------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <b>Inclusion</b>        | PSA $\leq 10$ , $\leq 3$ cores in one lobe, $\leq \text{T2a}$ , $\leq 3+4$ ; or all D'Amico groups     | Unilateral; Gleason $\leq 3+4$ to $\leq 4+3$ ; PSA $\leq 10$ –20; $\leq \text{T2}$ – $\text{T3a}$ ; some $<40$ mL gland, life expectancy $>5$ –10 years | Gleason 6–9, $\leq \text{cT3b}$ ; <b>65% intermediate, 28% high risk</b> |
| <b>Median follow-up</b> | 29–127 months                                                                                          | 6–39.5 months                                                                                                                                           | 32 months                                                                |
| <b>Oncological</b>      | Biochemical recurrence 26.6%; 5-year recurrence-free survival high, 10-year low; positive biopsy 41.7% | <b>Infield 6–54% (csPCa 5–21.4%); outfield 7–42.8% (csPCa 2–7.5%);</b> failure 11.9–31.2%                                                               | <b>7-year failure-free survival 69% (64–74%)</b>                         |
| <b>Sexual</b>           | About half maintained potency                                                                          | Mild fall in IIEF-5 or EPIC; ability to penetrate 50–78%                                                                                                | Not reported                                                             |
| <b>Complications</b>    | Retention 8.3%; some incontinence                                                                      | Transient dysuria and haematuria; debris passage; retention about 8%; urinary tract infection 6–18%; stricture 4%; RUF rare                             | Not reported                                                             |

- **What the table is for:** HIFU is the **second-largest FT literature** and the only modality with a **large prospective registry that includes high-risk disease**.
- **Pattern: infield failure persists into the MRI era.** Outfield csPCa is low, so the remaining problem is **ablation completeness** rather than selection.
- **Text and table differ.** For HEAT, the text reports failure-free survival **by risk group (68% intermediate, 65% high)**; the table gives the **overall 7-year figure (69%)**. For infield positive biopsy, the text quotes 16.7–54%, while individual series in the table run from 6%.

- **Trap:** transrectal HIFU is a **posterior-zone** tool. Anterior tumours are better reached transperineally or transurethraly.

### MRI-Guided Transurethral Ultrasound Ablation

- **In-bore MRI-guided focused ultrasound (MRgFUS):** MR thermometry confirms lethal temperature in the target and **safe temperatures at the urethra and sphincter** (Fig. 163.5). Phase II: a slight, non-significant EPIC urinary dip at 1 month that recovered by 3 months, with no change in other EPIC domains over 2 years.
- **MRI-guided transurethral ultrasound ablation (TULSA):** made possible by **miniaturised HIFU transducers** (Fig. 163.6). It combines **in-bore thermometry and anatomical resolution**, and the **transurethral route reaches the anterior prostate efficiently**.
- **Whole-gland TULSA phase I (multicentre):** safe; **spatial control of ablation within 1.3 mm**; no change in functional scores; grade 3 complications about 3%; **no grade 4–5**.
- **TULSA in low- and intermediate-risk disease:** about three-quarters free of biochemical progression and of salvage at 3 years, but **csPCa on 12-month biopsy in about a third**.

### Figure legends

- **Fig. 163.5 — MRgFUS** for Gleason 3+3 found on 26-core saturation biopsy: **T2 and DWI ( $b = 2000 \text{ s/mm}^2$ ) show a left posterior peripheral-zone lesion → intraoperative beam path and real-time thermal map → immediate post-contrast T1 shows non-perfused ablated volume → persists at 1 month → volume loss and fibrosis at 2 years** (Tay et al., 2017).
- **Fig. 163.6 — TULSA:** transurethral ultrasound applicator with an endorectal cooling device; thermal ablation boundary shown (Chin et al., 2016).

### Focal Salvage High-Intensity Focused Ultrasound

- In radiorecurrent disease (mostly intermediate or high risk before radiotherapy), **BPFS was 48% at 3 years and overall survival 92% at 5 years**.
- Complications: **urinary tract infection, bladder-neck stricture, urinary fistula (about 2%)**. About a quarter had **grade  $\geq 3$  events needing intervention**. **87% were continent at 2 years**.
- Focal salvage HIFU **offers benefit with relatively few complications** (Table 163.6). The **FORECAST trial** (ongoing) will compare outcomes of focal salvage cryotherapy and HIFU.

**TABLE 163.6 Summary of Published Focal HIFU Salvage Series**

Three series, 2012–2017.

| Feature                 | Range across series                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Selection</b>        | Biopsy-proven radiorecurrent, nonmetastatic; <b>TMB + mpMRI</b> or mpMRI + TRUS biopsy; about 36–42% high risk |
| <b>Ablation plan</b>    | Hemi-, quadrant, focal or index-lesion ablation                                                                |
| <b>Median follow-up</b> | 16–35 months                                                                                                   |
| <b>Oncological</b>      | <b>2-year BPFS/progression-free survival 49–52%; 3-year BPFS 48%</b>                                           |
| <b>Function</b>         | Mild IIEF-5 fall; <b>continence 64–87.5%</b>                                                                   |
| <b>Complications</b>    | Stricture, urinary tract infection, retention, <b>RUF (1–3 cases per series)</b> , osteitis                    |

- **What the table is for:** salvage HIFU controls only **about half at 2–3 years**, the same order as salvage cryotherapy.
- **Trap:** salvage continence (**64–87.5%**) is far below primary focal HIFU (**90–95%**).

## Focal Laser Ablation

- **Focal laser ablation (FLA):** coagulative necrosis from an **interstitially placed laser fibre**. Destruction is limited to a **fixed distance from the active fibre tip**, so the ablation zone is **predictable and can be used safely close to vital structures**.
- **Disadvantages:** **longer procedures or extra fibres for larger lesions**, and particular care is needed to obtain an adequate margin.
- **Histology:** **neovascularisation and increased tumour mitotic activity at the edge of the ablation zone** reinforce the need for a **safety margin** (Fig. 163.7).
- Fibre placed **transperineally or transrectally** via a delivery trocar.
- **In-bore MR-guided FLA technique** (phase II, Fig. 163.8): **14-gauge titanium introducer with polytetrafluoroethylene catheter into the lesion → 17-gauge laser applicator through each catheter → 60–120 s at 6–15 W, repeated as needed → target  $\geq 60^{\circ}\text{C}$ , with ablation extending  $>5$  mm beyond the lesion**. The cohort was mostly low risk; there was some infield and more outfield cancer at 12 months, with IIEF unchanged.
- **Office-based FLA under local anaesthesia** with a modified MRI-ultrasound fusion biopsy device (mainly Gleason 7) was safe but had **higher recurrence**, so **attention to the margin** is needed.
- The **longest follow-up (about 6 years)** showed a high failure-free rate but **infield recurrence in about 40%**. The authors called the infield recurrence and need for salvage **"disconcerting"** and urged close monitoring and better energy delivery.
- **Across series** (Table 163.7), over a median follow-up of 6–72.5 months: **continence 95–100%**; minimal change in IIEF or **Sexual Health Inventory for Men (SHIM)** scores; **infield recurrence 11–50%, outfield 16.7–60%**.
- **Severe adverse events are rare**. There was one grade III event by the **Common Terminology Criteria for Adverse Events (CTCAE)**, a urinary tract infection. Grade II events included **two RUFs, both healing with urinary catheterisation within 4–6 weeks**.

TABLE 163.7 Summary of Published Focal Laser Ablation Series

Seven series, 2010–2021.

| Feature                        | Range across series                                                                                                                                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Selection</b>               | mpMRI $\pm$ targeted/systematic biopsy; $\leq$ T2a–T2b; <b>PSA <math>&lt;10</math>–<math>15</math> ng/mL</b> ; PSA density $<0.15$ ; Gleason $\leq 7$ ( $\leq 3+4$ in most); lesion $\leq 15$ mm or $\leq 2$ mL in one series |
| <b>Ablation and monitoring</b> | Focal, often with a <b>5-mm margin</b> ; MRI thermometry, or contrast-enhanced ultrasound with fusion                                                                                                                         |
| <b>Median follow-up</b>        | 6–71.5 months                                                                                                                                                                                                                 |
| <b>Oncological</b>             | Infield positive biopsy 4% at 3 months to <b>50%</b> (csPCa 15–40%); outfield 16.7–60%; <b>failure-free survival 83% with 40% infield recurrence</b> at longest follow-up                                                     |
| <b>Function</b>                | IIEF unchanged or mild SHIM fall; continence 95–100%; International Prostate Symptom Score (IPSS) improved in one series                                                                                                      |
| <b>Complications</b>           | Haematuria (universal in one small series), perineal discomfort, ecchymosis, retention 8%, <b>RUF about 1.7%</b>                                                                                                              |

- **What the table is for:** laser is the most **"truly focal"** energy, yet it has the **smallest series, the shortest follow-up and the highest infield failure**.
- **Pattern:** a precise ablation zone does not guarantee cancer control. **A narrow zone means a narrow margin**.
- **Trap:** "predictable ablation zone = better oncology." In practice the edge of the ablation zone is where tumour survives.

### Figure legends

- **Fig. 163.7** — FLA for right transition-zone 3+4: **real-time MR temperature map → Arrhenius-based damage-estimation overlay → post-ablation contrast T1 showing non-enhancing necrosis → whole-mount histology with necrotic, perinecrotic and viable tumour zones** (Bomers et al., 2017).
- **Fig. 163.8** — MR-guided FLA equipment: **laser fibre in cooling catheter; perineal template with endorectal coil and titanium catheter; template grid overlaid on MRI; laser workstation** (Eggerer et al., 2016).

### Focal Irreversible Electroporation

- **Irreversible electroporation (IRE)** is **non-thermal**: **short pulses of direct current create irreversible membrane pores → cell death**.
- It claims to **preserve connective tissue architecture**, a possible scaffold for regrowth.
- **Ablation occurs between two or more probes placed 10–20 mm apart** (Fig. 163.9).
- **Delivery sequence**: **20 test pulses characterise the current between each probe pair → settings adjusted → deep anaesthesia to prevent muscle depolarisation → 70 pulses per probe pair for the final ablation. Field strength must exceed 1600 V/cm between every probe pair.**
- **Three early-phase studies** (about 100 men) used hemi- or anterior-zone ablation or template-grid focal ablation. They showed **no grade 4–5 complications** and **continence almost universally maintained**, with **some fall in erectile scores**.
- **Ablate-and-resect study** (resection at 1 month) found **fibrinoid necrosis of the unilateral neurovascular bundle and denudation of the prostatic urethra** in most patients. Whether the surviving connective tissue permits later regrowth and functional recovery is **unknown** (Fig. 163.10).
- **Larger cohorts** (three series, 451 men, mainly low and intermediate risk): **csPCa infield 1–9.8% and outfield 5–16%** at 6–12 months.
- **Matched-pair IRE vs robot-assisted radical prostatectomy (RARP)** (no head-to-head trial): **IRE significantly better for pad-free continence and erections sufficient for intercourse**, but some IRE patients had **residual csPCa requiring salvage**.

**TABLE 163.8 Summary of Published Focal Irreversible Electroporation Series**

Ten series, 2016–2023.

| Feature                 | Range across series                                                                                                                                    |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Selection</b>        | mpMRI + targeted and systematic biopsy or TMB; maximum Gleason <b>3+4 to 8</b> ; <T2b–T2c; PSA <15–20 ng/mL; some single-lesion only                   |
| <b>Ablation plan</b>    | Hemi-, anterior or template-based focal ablation                                                                                                       |
| <b>Median follow-up</b> | 6–60 months                                                                                                                                            |
| <b>Oncological</b>      | Infield positive biopsy 6–38.9% ( <b>csPCa 1–33.3%</b> ); outfield 5–16.7%; <b>failure-free survival 83% at 5 years</b> in the largest series          |
| <b>Urinary</b>          | <b>Pad-free 88–100%</b>                                                                                                                                |
| <b>Sexual</b>           | Erections sufficient for penetration about two-thirds (from about three-quarters at baseline); EPIC sexual domain falls; IIEF-5 unchanged to mild fall |
| <b>Complications</b>    | Mostly grade 1–2 (haematuria, dysuria, lower urinary tract symptoms, retention, urinary tract infection); stricture about 5%; <b>no grade 4–5</b>      |

- **What the table is for**: IRE's **functional results are its selling point**. Its oncological figures are of the same order as the thermal energies.
- **Trap**: "IRE spares the neurovascular bundle and urethra." Ablate-and-resect histology **shows injury to both**; the preservation claim is theoretical.

### Figure legends

- **Fig. 163.9 — IRE electrode metrics:** probes start periurethrally and are arranged radially; probe-pair spacing is 10–20 mm; each probe is **no more than 5 mm from capsule or urethra**; probes are parallel with **tip-to-capsule  $\geq 5$  mm**. Spacing (d) and exposed length (L) set the initial energy (Murray et al., 2016).
- **Fig. 163.10 — IRE ablate-and-resect:** perioperative TRUS with electrodes in place, and the matching whole-mount haematoxylin and eosin section (van den Bos et al., 2016).

### Focal Salvage Irreversible Electroporation

- In **biopsy-proven radio recurrent** disease (mostly low and intermediate risk; median follow-up 48 months), **progression-free survival was 60% and metastasis-free survival 91%**.
- Complications were mostly grade 1–2 (pain, urgency, mild haematuria). **Grade 3b: urethral sludge about 9%, rectal fistula about 1%.**
- **Continence 93%; erectile function preserved in 23%.**
- This is a single-institution series and needs multicentre confirmation (Table 163.9). The **SAFE pilot trial** (ongoing) is studying focal salvage IRE.

**TABLE 163.9 Summary of Published Focal Salvage IRE Series**

- A **single series (2023):** selection by **mpMRI + TMB + PET**; radio recurrent, nonmetastatic; **focal ablation**.
- Follow-up: PSA every 3 months, MRI at 6 months, TMB at 12 months.
- Outcomes: **progression-free survival 60%; continence 93%; retention 6%; RUF 1%;** perineal pain in a minority.
- **What the table is for:** IRE is the salvage option with the best reported continence, but the evidence rests on one institution.

**FOCAL SALVAGE MODALITIES COMPARED (RADIO RECURRENT DISEASE)**

|                               | Cryotherapy                          | HIFU                                                                                       | IRE                                                                   |
|-------------------------------|--------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Evidence</b>               | 8 series; largest literature         | 3 series                                                                                   | 1 series                                                              |
| <b>Oncological</b>            | BPFS 2-year 72–80%, 5-year about 50% | BPFS 2-year 49–52%, 3-year 48%                                                             | <b>Progression-free survival 60%, metastasis-free 91% (48 months)</b> |
| <b>Continence</b>             | 87–100%                              | <b>64–87.5%</b>                                                                            | 93%                                                                   |
| <b>Erections</b>              | 14–50% of potent men                 | Mild IIEF fall                                                                             | 23%                                                                   |
| <b>Signature complication</b> | RUF about 1.4%                       | <b>Bladder-neck stricture; fistula about 2%; grade <math>\geq 3</math> about a quarter</b> | <b>Urethral sludge about 9%; rectal fistula about 1%</b>              |
| <b>Ongoing trial</b>          | <b>FORECAST</b>                      |                                                                                            | <b>SAFE pilot</b>                                                     |

### Focal Vascular-Targeted Photodynamic (VTP) Therapy

- **Low-power near-infrared laser light** of a specific wavelength activates an **intravascular photosensitiser** preferentially taken up by tumour. **Reactive oxygen species** then damage cell membranes and small arterioles irreversibly, causing **tumour necrosis**. VTP is regarded as **both a drug and a device**.
- **TOOKAD Soluble (WST11, padeliporfin)**, a bacteriochlorophyll derivative: **highly soluble, stays within the vasculature even at high dose, and is rapidly cleared by the kidney and liver**.
- **Treatment sequence:** single intravenous dose of 4 mg/kg → transperineal intraprostatic fibre → 753-nm light at 150 mW/cm and 200 J/cm → about 2 hours of anaesthesia → **protective eyewear and low-level light for the following hour**.
- **Three phase II trials (Gleason 3+3):** safe. Infield positive biopsy was lower with **hemibliteration** than overall.

- **Randomised trial of focal VTP vs AS (Azzouzi, 2017) in 415 men with low-risk cancer:**
  - Significantly **more negative biopsies at 2 years, less progression and fewer men needing radical treatment.**
  - **Progression (volume, grade or PSA): 28% VTP vs 58% AS.**
  - **The first level 1 evidence that FT gives meaningful benefit over an accepted management strategy.**
  - **Limitations: Gleason 6 cohort only; short follow-up; surrogate end points** that do not reflect overall or cancer-specific survival; the outcome may represent **reclassification rather than true progression.**
- **Extended follow-up to 4 years, conversion to radical therapy (VTP vs AS): 2 years 7% vs 32%; 3 years 15% vs 44%; 4 years 24% vs 53%.** Progression was also reduced, and continence was well preserved, with **nil to mild IIEF fall at 24 months.**
- **VTP adverse events** were mostly mild to moderate and peri-procedural, resolving quickly. **Perineal pain** (from needle placement or prostatic swelling) affected **about 15% with VTP vs <1% with AS.**

**TABLE 163.10 Summary of Published Focal VTP Series**

Seven series, 2015–2019, including the combined analysis of three phase II trials and the randomised trial.

| Feature                 | Range across series                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Selection</b>        | <b>Gleason 3+3</b> (low-volume pattern 4 permitted in some); PSA ≤10–20 ng/mL; <T2a–T2c; mostly unilateral      |
| <b>Ablation plan</b>    | <b>Mostly hemiablation;</b> some subtotal or focal                                                              |
| <b>Median follow-up</b> | 12–68 months                                                                                                    |
| <b>Oncological</b>      | <b>Infield positive biopsy 19.4% (hemiablation) to 36.7%;</b> outfield 16.7–19%; csPCa infield 10%, outfield 6% |
| <b>Sexual</b>           | No significant to mild IIEF-5 fall; ability to penetrate 15.8% in one series                                    |
| <b>Complications</b>    | Haematuria, dysuria, perineal pain, haemospermia; urinary incontinence and urethral stenosis occasional         |

- **What the table is for:** VTP is the only modality with **randomised evidence**, but that evidence is in the group that **needs treatment least (Gleason 6).**
- **Randomised trial numbers:** the text reports **216 men randomised to VTP (196 treated, 195 followed to 24 months);** the source table lists 206. The two counts use different denominators.
- **Trap:** "VTP proven to improve survival." The end points were **progression and conversion**, not survival.

## Focal Radiofrequency Ablation

- **Radiofrequency ablation (RFA):** heat of **about 100 °C** from **molecular friction and agitation** denatures proteins and membranes, causing **coagulative necrosis.**
- **Generator up to 100 W, 450–1200 kHz;** probes are **monopolar or bipolar;** **thermocouples** give real-time temperature monitoring; control is **impedance- or temperature-based.**
- **Key advantage: limited conduction through fat,** which shields adjacent structures.
- **Bipolar focal RFA** (low and intermediate risk): csPCa on 6-month targeted biopsy in a fifth. At 12 months, most men were continent and able to penetrate. There were many adverse events, but **none CTCAE grade ≥3;** urinary obstruction resolved without sequelae.
- A second series found **infield csPCa in almost a third at 6 months,** with one grade 3 event (gross haematuria needing only diagnostic cystoscopy). Continence was preserved, but **IIEF fell significantly.** Technique and **margin planning** need to improve (Table 163.11).

**TABLE 163.11 Summary of Published RFA Series**

- **Two series (2020, 2021),** both selected by **mpMRI + TMB,** with Gleason 7 or ≤7 and PSA <10 ng/mL.

- **Infield positive biopsy 20–30%**; outfield 20% in one series; follow-up 6–12 months.
- Sexual: **ability to penetrate retained in 92%** in one series; **significant IIEF-5 fall** in the other.
- Complications: haematuria, haemospermia, urethral stricture; the large majority of events were grade 1–2.
- **What the table is for:** RFA is an **early-stage** technology with small numbers and short follow-up; margin planning is the lesson.

## Focal Brachytherapy

- **Treatment-simulation studies** show that brachytherapy can deliver an adequate focal dose.
- Its advantage is a **sharp fall-off in dose within a few millimetres of the source**. **High-dose-rate (HDR)** dose can be further sculpted by adjusting source dwell through the interstitial catheters.
- **Low-dose-rate (LDR, seed) brachytherapy vs HDR brachytherapy:** LDR urinary toxicity is **low**, **depending on baseline symptoms and urethral dose**. HDR carries a **urethral stricture rate up to 8.2% within 2 years**, at the **prostatic apex or bulbar urethra**.
- **Mid-term (4-year) focal HDR** in low- and intermediate-risk disease gave **poor results: biochemical disease-free survival 70%, outfield csPCa in a third, and deterioration to moderate or severe erectile dysfunction in half**. There was no grade 2 genitourinary or gastrointestinal toxicity. The authors blamed **poor patient selection**.
- **Unique follow-up: mandatory biopsy at 24 months** in published series. This delays confirmation of success and makes **early comparison with non-radiation FT difficult**.

**TABLE 163.12 Summary of Published Focal Brachytherapy Series**

Seven series, 2017–2020; rows split by dose rate.

| Feature                 | LDR (seed) series                                                                               | HDR series                                                                        |
|-------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Selection</b>        | mpMRI + TMB or targeted/systematic biopsy; Gleason $\leq 3+4$ ; PSA $\leq 10$ ng/mL; $\leq T2a$ | mpMRI $\pm$ fusion biopsy; Gleason $\leq 7$ ; PSA $\leq 10$ –15 ng/mL; $\leq T2c$ |
| <b>Ablation plan</b>    | Focal or hemiablation                                                                           | Focal or hemiablation                                                             |
| <b>Median follow-up</b> | 12–24 months                                                                                    | 6–48 months                                                                       |
| <b>Follow-up biopsy</b> | <b>12 or 24 months</b>                                                                          | 12 months, or for cause / PET-CT after biochemical recurrence                     |
| <b>Oncological</b>      | No recurrence to <b>19% infield / 8% outfield</b> ; csPCa outfield 6%                           | No recurrence (at 6 months) to <b>33% outfield</b> ; 14% biochemical recurrence   |
| <b>Sexual</b>           | No significant change; <b>73% preserved ability to penetrate</b>                                | <b>50% new impotence</b> ; 34% preserved; mild IIEF fall                          |
| <b>Toxicity</b>         | None reported                                                                                   | <b>No grade 2 genitourinary or gastrointestinal toxicity</b> ; dysuria, cystitis  |

- One further series (dose rate unspecified; Gleason  $\leq 3+4$ , PSA  $\leq 15$  ng/mL) reported a rise in IPSS and a mild IIEF-5 fall.
- **What the table is for:** focal brachytherapy is **functionally mild but oncologically unproven**, and HDR has the **worst erectile outcome** of the focal series.
- **Biopsy timing:** the text describes **mandatory biopsy at 24 months** as the rule, but several series in the table biopsied at **12 months**. The text describes the typical protocol; the table shows its variants.
- **Trap:** radiation toxicity appears **late**, so short follow-up flatters it.

## Other Modalities

### Focal Intensity-Modulated Radiation Therapy

- External-beam advances allow a **highly focused intraprostatic dose that spares vital structures**.
- **Randomised phase II trial, urethral-sparing intensity-modulated radiation therapy (IMRT) vs standard IMRT (low risk): no genitourinary toxicity benefit, and worse oncological outcome** at median 4.7 years.
- In contrast, another series reported good 3-year BPFS across low, intermediate and high risk, with mostly grade 1 genitourinary toxicity.
- **Focal stereotactic body radiotherapy (SBRT)**: dosimetric feasibility has been shown but there is no clinical trial. A five-patient pilot showed **better dosimetry, minimal toxicity and an early favourable PSA response**. The results are preliminary, and **radiation toxicity appears late**, so long-term multicentre data are needed.

### Focal Nanoparticles

- **Photothermal excitation of nanoshells that accumulate preferentially in the fenestrated perivascular spaces of tumours** → preferential cancer-cell death. A drug and device therapy.
- An open-label multicentre trial of a single dose of **AuroLase** nanoparticle-directed focal excitation (low to intermediate risk) reported **about three-quarters success on 12-month biopsy**.
- **PRX302**: a **genetically modified pore-forming protein (aerolysin) activated by enzymatically active PSA** within the prostate. In a pilot study a single MR-fusion transperineal dose was **well tolerated**, with some complete and some partial histological responses.
- **NanoPac**: sterile **paclitaxel nanoparticles** given by intratumoral injection (phase IIa). In men with **Gleason  $\geq 7$** , it was injected under TRUS guidance with mpMRI into the IL and its lobe, followed by radical prostatectomy at 4 weeks.
  - The proportion of adenocarcinoma fell in some specimens. At **15 mg/mL**, lesion volume on mpMRI and PSA density both fell.
  - **No drug-related serious adverse events**, prostatitis or dose-limiting toxicity occurred, so the dose was escalated to 15 mg/mL.
  - **Paclitaxel persisted in the prostate and lymph nodes more than 30 days** after injection.

### Adjuvants to Focal Therapy

- Many proponents think successful FT will rest on a **combination of device and adjunct**, not on a device alone.
- **SBRT with a boost to the MR-visible tumour (5–10 fractions): no biochemical recurrence at 2 years** in a pilot study; acute genitourinary/gastrointestinal toxicity 8%/2%; late grade 3 toxicity 6%/0%. The added value of **non-radiation focal ablation plus low-dose external beam** is undetermined.
- **DOTA-AR**, a bombesin-antagonist peptide targeting the **gastrin-releasing peptide receptor (GRPr)**, conjugated to a radioisotope and combined with WST11, delayed tumour growth in **GRPr-positive xenografts**.

### CRYOABLATION ENHANCERS INVESTIGATED

| Adjunct                                                         | Finding                                                                                                    |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Androgen deprivation                                            | No useful increase in cell death                                                                           |
| Solutions with a higher-than-physiological eutectic temperature | Promote extra- and intracellular <b>eutectic ice</b> , adding a kill mechanism to intracellular lethal ice |

| Adjunct                                    | Finding                                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chemotherapeutic apoptosis inducers</b> | 5-fluorouracil, cisplatin, doxorubicin, mitomycin, bleomycin, vinorelbine, TNF-related apoptosis-inducing ligand: <b>more cell death at milder temperatures in vitro</b> |
| <b>TNF-<math>\alpha</math></b>             | Proinflammatory cytokine being explored as an enhancer                                                                                                                   |
| <b>Vitamin D3</b>                          | Improves apoptosis at <b>sublethal isotherms</b> in murine models                                                                                                        |

- **No widely applicable substance yet** augments device-mediated cell kill.

**Concept** — Every energy source fails **at the edge of the ablation zone**, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants are all attempts to make that edge lethal.

#### FOCAL ENERGY MODALITIES COMPARED

| Modality             | Mechanism / lethal parameter                                         | Route                           | Monitoring                              | Strength                                                 | Weakness                                                                      |
|----------------------|----------------------------------------------------------------------|---------------------------------|-----------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Cryotherapy</b>   | Heat extraction; ice, ischaemia, apoptosis                           | Transperineal                   | <b>Visible ice edge</b> ; thermocouples | Largest literature; registry comparison; urethral warmer | Retention; RUF                                                                |
| <b>HIFU</b>          | Focused ultrasound, $\geq 55^{\circ}\text{C}$                        | Transrectal                     | Ultrasound (or MR thermometry)          | Posterior lesions; short focal time; HEAT registry       | <b>Anterior lesions</b> ; gland size, height, calcification                   |
| <b>TULSA</b>         | Transurethral ultrasound                                             | Transurethral                   | <b>MR thermometry, within 1.3 mm</b>    | Anterior gland                                           | In-bore MRI resource                                                          |
| <b>Laser</b>         | Interstitial fibre, $\geq 60^{\circ}\text{C}$                        | Transperineal or transrectal    | MR thermometry                          | Most predictable zone                                    | <b>Highest infield failure</b> ; extra fibres for large lesions               |
| <b>IRE</b>           | Non-thermal pores, $>1600\text{ V/cm}$                               | Transperineal                   | Probe geometry                          | <b>Best function vs RARP</b>                             | Needs deep anaesthesia; neurovascular bundle and urethral injury on histology |
| <b>VTP</b>           | Photosensitiser + 753-nm light $\rightarrow$ reactive oxygen species | Transperineal fibre             | —                                       | <b>Only randomised evidence</b>                          | Low-risk only; light precautions                                              |
| <b>RFA</b>           | About $100^{\circ}\text{C}$ , molecular friction                     | Interstitial probes             | Thermocouples                           | Fat limits conduction                                    | Early data only                                                               |
| <b>Brachytherapy</b> | Radiation; sharp dose fall-off                                       | Interstitial catheters or seeds | Dosimetry                               | Mild early toxicity (LDR)                                | Late toxicity; 24-month biopsy; HDR stricture and erectile loss               |

#### PRIMARY FOCAL THERAPY OUTCOMES COMPARED

| Modality           | Follow-up   | Infield positive biopsy                   | Outfield positive biopsy | Continence | Erectile function |
|--------------------|-------------|-------------------------------------------|--------------------------|------------|-------------------|
| <b>Cryotherapy</b> | 6–85 months | 7.1–38.9% overall (about a quarter csPCa) |                          | 95.1–100%  | 46.8–100%         |
| <b>HIFU</b>        | 6–45 months | 16.7–54% (csPCa 0–21.8%)                  | 7–42.8% (csPCa 2–7.5%)   | 90–95%     | 50–78%            |

| Modality      | Follow-up                    | Infield positive biopsy | Outfield positive biopsy | Continence  | Erectile function                     |
|---------------|------------------------------|-------------------------|--------------------------|-------------|---------------------------------------|
| Laser         | 6–72.5 months                | 11–50%                  | 16.7–60%                 | 95–100%     | Minimal change                        |
| IRE           | 6–12 months (larger cohorts) | csPCa 1–9.8%            | csPCa 5–16%              | Almost all  | Some decline                          |
| VTP           | 12–68 months                 | 19.4–36.7%              | 16.7–19%                 | Preserved   | Nil to mild IIEF fall                 |
| RFA           | 6–12 months                  | csPCa 20–30%            | 20%                      | Most to all | 92% penetrate / significant IIEF fall |
| Brachytherapy | 6–48 months                  | 0–19%                   | Up to 33%                | —           | 34–73% preserved                      |

- **Pattern: continence is preserved by every modality** (about 90–100%). The modalities differ in **infield failure**, and the heterogeneous definitions forbid ranking.
- **Trap:** comparing modalities across series. Selection, biopsy protocol and end-point definitions all differ, and only VTP has a randomised comparison.

## POST-FOCAL THERAPY FOLLOW-UP

- **PSA is less informative after FT.** It should be **undetectable after radical prostatectomy**, and **after radiotherapy a defined rise** is needed for biochemical recurrence. **After FT the fall is variable, 50–80%**, because **residual epithelium keeps producing PSA**.
- Proposed thresholds: **nadir + 1.0 ng/mL at 12 months**, or **nadir + 1.5 ng/mL at 24–36 months**, gave high sensitivity and specificity for csPCa on for-cause biopsy.
- **Digital examination is unreliable** because of scarring and contraction of the tumour site. On mpMRI, TRUS and **contrast-enhanced ultrasound (CEUS)**, the **ablation zone first enlarges and then the gland shrinks** through fibrosis and deformation.
- **2015 International Consultation on Urological Diseases (ICUD)–Société Internationale d'Urologie (SIU) definition of success:**

### DEFINING SUCCESS AFTER FOCAL THERAPY (ICUD–SIU, 2015)

|                           | Infield (treated area)                          | Outfield (untreated area)                                        |
|---------------------------|-------------------------------------------------|------------------------------------------------------------------|
| Short term                | Eradication of the treated tumour focus or foci | Absence of csPCa. <b>Any outfield cancer = selection failure</b> |
| Intermediate to long term | Absence of csPCa                                | Absence of csPCa. <b>Outfield cancer may be de novo</b>          |

- **Recommended schedule:** mpMRI + **mandatory targeted biopsy (4–6 cores) of the treated area at 3–6 months** → mpMRI at 1–2 and 5 years, with **targeted biopsy as needed (especially for a new lesion)** → **systematic biopsy at 1–2 and 5 years** to assess the untreated zone when low-volume, low-grade disease is being monitored.
- **Post-FT mpMRI experience is limited.** After whole-gland HIFU, DCE-MRI sensitivity is 73–87% and specificity 73–82%.
- **Prostate Imaging after Focal Ablation (PI-FAB)** reads mpMRI sequences **in a fixed order** (Fig. 163.11):
  - **DCE is read first**, because scarring and fibrosis obscure the T2-weighted and DWI sequences that PI-RADS relies on.
  - **Enhancement >3 mm within the treated zone is an indication for biopsy.**

- The criteria are expert derived, so clinical correlation is still advised until validation. PI-FAB is a **step towards standardised post-FT imaging**.
- To reduce the cost of mpMRI and the morbidity of repeated biopsy, **CEUS was as good as mpMRI** for early ablation success after focal IRE, whereas **grey-scale TRUS cannot see the ablation zone**.

#### Figure legend

- **Fig. 163.11 — PI-FAB scoring:** read sequences in order **DCE → DWI (high b value) → DWI (apparent diffusion coefficient, ADC, map) → T2-weighted**. A positive finding is focal early enhancement on DCE, focal hyperintensity at high b value, and focal hypointensity on the ADC map and T2.

| PI-FAB | DCE      | DWI (high b) | Management                                                                                                          |
|--------|----------|--------------|---------------------------------------------------------------------------------------------------------------------|
| 1      | Negative | Negative     | <b>Continue monitoring</b>                                                                                          |
| 2      | Positive | Negative     | <b>Correlate with PSA kinetics</b> (and follow-up MRI); biopsy if PSA rising. Compare DCE with pretreatment imaging |
| 3      | Positive | Positive     | <b>Biopsy</b>                                                                                                       |

- **Downgrade 2 → 1** if the enhancement is **linear and not at the site of the original tumour**.
- **Upgrade 2 → 3** for **early focal enhancement >3 mm resembling the original tumour**, or a PI-FAB 2 focus that has grown.

**Concept** — After FT there is no PSA "zero" to aim for. Success is defined **anatomically (infield vs outfield) and proven by MRI-directed biopsy**, and because scar hides T2 and DWI changes, **DCE is read first**.

## CONCLUSIONS AND AREAS FOR FUTURE RESEARCH

- **Willet Whitmore: "Is cure possible when necessary and necessary when possible?"**
- **Longer life expectancy, awareness and PSA screening** mean more cancers are diagnosed, and better imaging and biopsy find them earlier. In a subset, FT may **cure or control the cancer for a long time**. Given its slow biology, prostate cancer can be **converted from an acute to a chronic disease**.
- **FT + AS of the untreated gland avoids overtreatment and preserves genitourinary function**. Many men value quality of life as highly as cancer control, and image-guided ablation can deliver both. It remains a **work in progress**.
- **Three open problems:**
  1. **Can untreated satellite "insignificant" cancers safely be left?** The answer is unknown, a question AS shares. Small cancers may progress, and the **randomised VTP trial's progression-free benefit in low-risk disease** suggests value in treating lower-grade lesions. **Clinical, imaging and genetic biomarkers** are needed, possibly with functional and metabolic imaging. Meanwhile, **close surveillance continues**.
  2. **Long-term cost-effectiveness is unknown. Up to half of men on AS eventually have radical treatment**, and many FT patients will need further focal or radical treatment. Patients must accept possible **repeat treatment**. The cost and morbidity of repeated FT, and of radical treatment after FT, are undefined.
  3. **Adjuvants may be needed**. Wide excision for breast cancer is routinely paired with **adjuvant radiotherapy**, but **no prostate trial has tested focal ablation with whole-gland irradiation**. Several adjuvants are in early development.

**Concept** — The breast analogy that launched FT also shows its gap. **Lumpectomy succeeds because the rest of the organ receives adjuvant treatment**; prostate FT instead relies on surveillance and retreatment.